

Inverse of 2×2 Matrices

Linear Algebra Worksheet · Grade 10–12

Name: _____

Date: _____

Learning Objectives

- Verify that two matrices are inverses by showing their product equals the identity matrix
- Apply the inverse formula to find the inverse of a 2×2 matrix using the determinant
- Identify when a 2×2 matrix is not invertible (singular matrix)

Problems

1. Identify the 2×2 identity matrix from the list below. Which of the following is the identity matrix?

2. Find the determinant of matrix A.

$$\begin{vmatrix} 4 & 2 \\ 3 & 1 \end{vmatrix}$$

3. Multiply matrix A times matrix B and verify that the result is the identity matrix.

$$\begin{bmatrix} 2 & 1 \\ 5 & 3 \end{bmatrix}$$

4. Find the inverse of matrix A using the inverse formula.

$$\begin{bmatrix} 3 & 1 \\ 2 & 1 \end{bmatrix}$$

5. Find the inverse of matrix A using the inverse formula.

$$\begin{bmatrix} 5 & 2 \\ 3 & 1 \end{bmatrix}$$



6. Determine whether the matrix below is invertible. If it is not invertible, explain why.

$$\begin{bmatrix} 2 & 4 \\ 1 & 2 \end{bmatrix}$$

7. Show that B is the inverse of A by proving that A times B equals the identity matrix AND B times A equals the identity matrix.

$$\text{A} = \begin{bmatrix} 1 & 1 \\ -2 & -1 \end{bmatrix}, \quad \text{B} = \begin{bmatrix} -1 & -1 \\ 2 & 1 \end{bmatrix}$$

8. Find the inverse of matrix A, then verify your answer by computing A times A-inverse.

$$\begin{bmatrix} 4 & 7 \\ 2 & 3 \end{bmatrix}$$

9. Matrix A is given below. Find A-inverse, then use it to solve the system of equations for x and y by computing A-inverse times the column vector [8, 5].

$$\begin{bmatrix} 3 & 2 \\ 4 & 3 \end{bmatrix}$$

10. Given that matrix A equals the one shown, find the value of k such that the matrix is NOT invertible (i.e., its determinant equals zero), then find the inverse of A when k equals 1.

$$\begin{bmatrix} k & 3 \\ 2 & 6 \end{bmatrix}$$



Inverse of 2×2 Matrices — Answer Key

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Answer Key

1. Answer: The matrix with 1s on the main diagonal and 0s elsewhere

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

- The identity matrix I has 1s along the main diagonal (top-left to bottom-right).
- All other entries are 0.
- So the 2×2 identity matrix is $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$.

2. Answer: $\det(A) = -2$

- For a 2×2 matrix $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$, the determinant is $ad - bc$.
- $\det(A) = (4)(1) - (2)(3)$
- $\det(A) = 4 - 6 = -2$

3. Answer: $A \times B = I$ (identity matrix)

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

- Multiply row 1 of A by column 1 of B : $(2)(3) + (1)(-5) = 6 - 5 = 1$
- Multiply row 1 of A by column 2 of B : $(2)(-1) + (1)(2) = -2 + 2 = 0$
- Multiply row 2 of A by column 1 of B : $(5)(3) + (3)(-5) = 15 - 15 = 0$
- Multiply row 2 of A by column 2 of B : $(5)(-1) + (3)(2) = -5 + 6 = 1$
- Result: $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I$ ✓

4. Answer: A inverse = $\begin{bmatrix} 1 & -1 \\ -2 & 3 \end{bmatrix}$

$$\begin{bmatrix} 1 & -1 \\ -2 & 3 \end{bmatrix}$$

- Identify $a=3$, $b=1$, $c=2$, $d=1$.
- Calculate the determinant: $\det = ad - bc = (3)(1) - (1)(2) = 3 - 2 = 1$.
- Apply the inverse formula: $A^{-1} = (1/\det) \times \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$.
- $A^{-1} = (1/1) \times \begin{bmatrix} 1 & -1 \\ -2 & 3 \end{bmatrix}$
- $A^{-1} = \begin{bmatrix} 1 & -1 \\ -2 & 3 \end{bmatrix}$

5. Answer: A inverse = $\begin{bmatrix} -1 & 2 \\ 3 & -5 \end{bmatrix}$



$$\begin{bmatrix} -1 & 2 \\ 3 & -5 \end{bmatrix}$$

- Identify $a=5$, $b=2$, $c=3$, $d=1$.
- Calculate the determinant: $\det = (5)(1) - (2)(3) = 5 - 6 = -1$.
- Apply the inverse formula: $A^{-1} = (1/-1) \times \begin{bmatrix} 1 & -2 \\ -3 & 5 \end{bmatrix}$.
- $A^{-1} = \begin{bmatrix} -1 & 2 \\ 3 & -5 \end{bmatrix}$

6. Answer: Not invertible; determinant = 0

- Calculate the determinant: $\det = (2)(2) - (4)(1) = 4 - 4 = 0$.
- Because the determinant equals 0, the matrix is singular (not invertible).
- A matrix is only invertible when its determinant is non-zero.

7. Answer: Both AB and BA equal the identity matrix, so B is the inverse of A.

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

- Compute AB: $\text{row1} \times \text{col1} = (1)(-1) + (1)(2) = 1$, $\text{row1} \times \text{col2} = (1)(-1) + (1)(1) = 0$
- $\text{row2} \times \text{col1} = (-2)(-1) + (-1)(2) = 0$, $\text{row2} \times \text{col2} = (-2)(-1) + (-1)(1) = 1 \rightarrow AB = I \checkmark$
- Compute BA: $\text{row1} \times \text{col1} = (-1)(1) + (-1)(-2) = 1$, $\text{row1} \times \text{col2} = (-1)(1) + (-1)(-1) = 0$
- $\text{row2} \times \text{col1} = (2)(1) + (1)(-2) = 0$, $\text{row2} \times \text{col2} = (2)(1) + (1)(-1) = 1 \rightarrow BA = I \checkmark$
- Since $AB = BA = I$, B is the inverse of A.

8. Answer: A inverse = $\begin{bmatrix} -3 & 7 \\ 2 & -4 \end{bmatrix}$

$$\begin{bmatrix} -3 & 7 \\ 2 & -4 \end{bmatrix}$$

- Identify $a=4$, $b=7$, $c=2$, $d=3$.
- $\det = (4)(3) - (7)(2) = 12 - 14 = -2$.
- $A^{-1} = (1/-2) \times \begin{bmatrix} 3 & -7 \\ -2 & 4 \end{bmatrix} = \begin{bmatrix} -3/2 & 7/2 \\ 1 & -2 \end{bmatrix}$.
- Verify: $A \times A^{-1}$ $\text{row1} \times \text{col1} = (4)(-3/2) + (7)(1) = -6 + 7 = 1 \checkmark$
- $\text{row1} \times \text{col2} = (4)(7/2) + (7)(-2) = 14 - 14 = 0 \checkmark$; similarly rows 2 gives $[0, 1] \checkmark$

9. Answer: $x = 14$, $y = -17$

- $\det = (3)(3) - (2)(4) = 9 - 8 = 1$.
- $A^{-1} = (1/1) \times \begin{bmatrix} 3 & -2 \\ -4 & 3 \end{bmatrix} = \begin{bmatrix} 3 & -2 \\ -4 & 3 \end{bmatrix}$.
- Multiply A^{-1} by $\begin{bmatrix} 8 & 5 \end{bmatrix}$: $x = (3)(8) + (-2)(5) = 24 - 10 = 14$
- $y = (-4)(8) + (3)(5) = -32 + 15 = -17$
- Solution: $x = 14$, $y = -17$

10. Answer: $k = 1$ makes $\det = 0$... wait: not invertible when $k = 1$; inverse when $k = 2$ is $(1/6)\begin{bmatrix} 6 & -3 \\ -2 & 2 \end{bmatrix}$

- $\det = 6k - (3)(2) = 6k - 6$.





- Now find inverse when $k = 2$: matrix is $\begin{bmatrix} 2 & 3 \\ 2 & 6 \end{bmatrix}$, $\det = (2)(6) - (3)(2) = 12 - 6 = 6$.
 - $A^{-1} = (1/6) \times \begin{bmatrix} 6 & -3 \\ -2 & 2 \end{bmatrix}$
 - $A^{-1} = \begin{bmatrix} 1 & -1/2 \\ -1/3 & 1/3 \end{bmatrix}$
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